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Strengthening Career and Technical Education for the 21st Century Act (Perkins V): A Primer

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SUMMARY

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Strengthening Career and Technical Education for the 21st Century Act (Perkins V): A Primer

The Carl D. Perkins Career and Technical Education Act (Perkins Act) is the primary federal law aimed at developing and supporting career and technical education (CTE) programs for secondary and postsecondary students. In the 109th Congress, the Perkins Act was reauthorized by the Carl D. Perkins Career and Technical Education Improvement Act of 2006 (Perkins IV; P.L. 109-270). The Perkins Act was most recently amended in the 115th Congress by the Strengthening Career and Technical Education for the 21st Century Act (Perkins V; P.L. 115-224). Perkins V was signed into law by President Trump in July 2018 and went into effect on July 1, 2019. Appropriations are authorized for FY2019 through FY2024.

The purpose of the act is to develop more fully the academic knowledge and technical and employability skills of secondary and postsecondary students who elect to enroll in CTE programs. More than $1.3 billion was appropriated in FY2021 for Perkins V.

The largest program authorized by Perkins V is the Basic State Grants program. Key features of this program include the following:

- formula grants to the states that are allocated based on the age distribution of a state's population and state per capita income;
- a distribution of at least 85% of the funds from each state to the local level, with no more than 10% reserved for state leadership activities (such as recruiting CTE teachers) and not more than 5% used for state administration;
- federally approved state plans to design, implement, and support CTE programs tailored to each state's economic and workforce development needs;
- state flexibility in deciding the allocation of state funds between secondary and postsecondary local CTE providers;
- requirements for states to develop and implement programs of study, which are nonduplicative sequences of courses that lead to an industry-recognized credential, certificate, or degree;
- core indicators of performance for accountability purposes, with target levels of performance established by each state;
- disaggregation of performance data by special populations (e.g., individuals with disabilities or from economically disadvantaged families) and subgroups (e.g., races and ethnicities); and
- a requirement for states and local providers to prepare and implement program improvement strategies if the target levels on core indicators of performance are not met.

Perkins V also authorizes funding for National Activities and for Tribally Controlled Postsecondary Career and Technical Institutions, with funding at approximately $8 million and $10 million in FY2021, respectively. National Activities includes an Innovation and Modernization (I&M) competitive grant program to study and replicate evidence-based practices.

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Introduction

The Carl D. Perkins Career and Technical Education Act (Perkins Act), as amended most recently by the Strengthening Career and Technical Education for the 21st Century Act (Perkins V; P.L. 115-224), is the primary federal law supporting career and technical education (CTE) services offered within the states[1] at the secondary and postsecondary education levels.[2] CTE, once known as vocational education, consists of educational activities that provide individuals with the knowledge and skills needed to prepare for specific careers in current or emerging professions. CTE also prepares students for the labor market more broadly by integrating CTE with academic standards. For a brief history of Perkins Acts, see **Appendix A**.

Appropriations for Perkins V were authorized by statute for FY2019 through FY2024.[3] It was funded at $1.3 billion for FY2021.[4] The statutory purpose of Perkins V is to support the development of career and technical, as well as academic, skills among secondary and postsecondary education students enrolled in CTE programs. It is unique among federal education programs with its emphasis on in-demand industry sectors or occupations[5] paired with specific programs of study that lead to an industry-recognized degree or credential.

This report provides an overview of Perkins V, including its design, implementation, and funding as described in statute. After a brief introduction to CTE, the report is divided into sections that focus on a detailed description of the Basic State Grants (BSG), discuss National Activities and other programs under Title I, and summarize additional provisions under Title II. A final section summarizes Perkins V authorizations and appropriations. The appendices provide a brief history of federal involvement in CTE as well as formulas and allocations for the BSG.

Career and Technical Education under Perkins V

For the purposes of Perkins V, CTE includes organized educational activities that impart technical or occupational skills and academic content at the secondary and postsecondary levels and lead to an industry-recognized credential, a certificate, or an associate's degree, or to further education. CTE courses are more directly tied to specific professions or occupational fields than liberal arts courses.[6]

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[1] A *state* is defined in Perkins V to refer to the 50 U.S. states, the District of Columbia, the Commonwealth of Puerto Rico, the U.S. Virgin Islands, Guam, American Samoa, the Commonwealth of the Northern Mariana Islands, and the Republic of Palau.

[2] For additional information on the history of federal involvement in CTE, please refer to CRS Report R44542, *Carl D. Perkins Career and Technical Education Act of 2006: An Overview*.

[3] Though Perkins V is authorized through FY2024, the act is automatically extended through FY2025 by the General Education Provisions Act (GEPA). For more information on GEPA's contingent extension of programs, see pages 3-4 of CRS Report R41119, *General Education Provisions Act (GEPA): Overview and Issues.*

[4] U.S. Department of Education, Department of Education Budget Tables, *FY 2021 Congressional Action*, available at https://www2.ed.gov/about/overview/budget/tables.html?src=ct.

[5] Perkins V uses the definition of *in-demand industry sector or occupation* as defined in Section 3 of the Workforce Innovation and Opportunity Act (WIOA), which is "(i) an industry sector that has a substantial current or potential impact (including through jobs that lead to economic self-sufficiency and opportunities for advancement) on the state, regional, or local economy, as appropriate, and that contributes to the growth or stability of other supporting businesses, or the growth of other industry sectors; or (ii) an occupation that currently has or is projected to have a number of positions (including positions that lead to economic self-sufficiency and opportunities for advancement) in an industry sector so as to have a significant impact on the state, regional, or local economy, as appropriate."

[6] Liberal arts courses include General Studies and Humanities; Social Sciences; English Language and
(continued...)

Perkins V Definition of Career and Technical Education

Under Perkins V, *career and technical education* is defined as "organized educational activities that

(A) offer a sequence of courses that—

(i) provides individuals with rigorous academic content and relevant technical knowledge and skills needed to prepare for further education and careers in current or emerging professions, which may include high-skill, high-wage or in-demand industry sectors or occupations, which shall be, at the secondary level, aligned with the challenging State academic standards adopted by a State under section 1111(b)(1) of the Elementary and Secondary Education Act (ESEA) of 1965;

(ii) provides technical skill proficiency or a recognized postsecondary credential, which may include an industry-recognized credential, a certificate, or an associate degree; and

(iii) may include prerequisite courses (other than a remedial course) that meet the requirements of this subparagraph; and

(B) include competency-based, work-based, or other applied learning that supports the development of academic knowledge, higher-order reasoning and problem-solving skills, work attitudes, employability skills, technical skills, and occupation-specific skills, and knowledge of all aspects of an industry, including entrepreneurship, of an individual." (§7(5)).

Career and Technical Education at the Secondary Education Level

CTE at the secondary level consists of programs designed to prepare students for specific careers and occupational fields. Completing a secondary CTE program may lead to an industry-recognized credential or certificate, continuation of education at the postsecondary level, or immediate entry into the workforce.

Secondary CTE providers typically include

- public and private high schools, including Bureau of Indian Education (BIE) schools;[7]
- CTE-specific schools, area CTE schools,[8] and career academies within comprehensive high schools, which are all focused on providing occupational preparation;

Literature/Letters; Theology and Religious Vocations; History; Area, Ethnic, Cultural, Gender, and Group Studies; Foreign Languages, Literatures, and Linguistics; and Philosophy and Religious Studies as classified in the Integrated Postsecondary Education Data System (IPEDS).

[7] The BIE within the U.S. Department of Interior funds a system of elementary, secondary, and postsecondary schools for Indian students. For more information, see CRS Report RL34205, *Indian Elementary-Secondary Education: Programs, Background, and Issues*.

[8] An area CTE school is

(A) a specialized public secondary school used exclusively or principally for the provision of career and technical education to individuals who are available for study in preparation for entering the labor market; (B) the department of a public secondary school exclusively or principally used for providing career and technical education in not fewer than 3 different fields that are available to all students, especially in high-skill, high-wage, or in-demand industry sectors or occupations; (C) a public or nonprofit technical institution or career and technical education school used exclusively or principally for the provision of career and technical education to individuals who have completed or left secondary school and who are available for study in preparation for entering the labor market, if the institution or school admits, as regular students, individuals who have completed secondary school and individuals who have left secondary school; or (D) the department or division of an institution of higher education, that operates under the policies of the eligible agency and that provides career and technical education in not fewer than 3 different occupational fields leading to immediate employment but not necessarily leading to a baccalaureate degree, if

(continued...)

- detention centers and correctional facilities; and
- cooperative programs with community or technical colleges.

CTE courses are widely available to high school students. According to the latest data, published by the National Center for Education Statistics in 2020, the majority (88%) of public high school graduates in 2013 attained at least one CTE credit and are considered CTE participants; 38% earned at least two CTE credits in a single occupational area,[9] satisfying the Perkins V definition of a CTE concentrator.[10]

Career and Technical Education at the Postsecondary Education Level

CTE at the postsecondary level consists of programs designed to prepare students for specific careers and occupational fields. Completing a postsecondary CTE program may lead to an associate's degree, an industry-recognized credential, or a certificate.

Postsecondary CTE providers typically include

- community and technical colleges;
- private two-year colleges;
- public and private four-year universities;
- vocational schools;
- employers, labor organizations, and other industry organizations that provide apprenticeship and pre-apprenticeship programs;
- adult workforce education centers; and
- correctional facilities.

Perkins V

Perkins V (P.L. 115-224) was enacted on July 31, 2018, and provided authorizations of appropriations for several programs for FY2019 through FY2024.

Structure and Purpose of the Act

Perkins V authorizes funding for three separate programs or initiatives:

- the BSG program;[11]

the department or division admits, as regular students, both individuals who have completed secondary school and individuals who have left secondary school." (Perkins V, §7(3))

[9] Albert Liu and Laura Burns, *Public High School Students' Career and Technical Education Coursetaking: 1992 to 2013,"* National Center for Education Statistics, Institute of Educational Science, https://nces.ed.gov/pubs2020/2020010.pdf.

[10] The terms *CTE participant* and *CTE concentrator* are used throughout Perkins V. A participant means an individual who completes not less than one course in a CTE program or program of study at either the secondary or postsecondary level. A concentrator at the secondary level is a student who has completed at least two courses in a single CTE program or program of study. A concentrator at the postsecondary level is a student who has earned at least 12 credits within a CTE program or program of study, or who has completed a program if the program encompasses fewer than 12 credits or the equivalent in total.

[11] Perkins V, Title I.

- National Activities;[12] and
- the Tribally Controlled Postsecondary Career and Technical Institutions (TCPCTI) Program.[13]

Perkins V provides the main source of direct federal funding for CTE programs.[14] The purpose of the act is to develop more fully the academic knowledge and employability skills of secondary and postsecondary education students by[15]

- building on the efforts of states and localities to develop challenging academic and technical standards and to assist students in meeting such standards, including preparation for high-skill, high-wage, or in-demand occupations in current or emerging professions;
- promoting the development of services and activities that integrate rigorous and challenging academic and career and technical instruction, and that link secondary education and postsecondary education for participating CTE students;
- increasing state and local flexibility in providing services and activities designed to develop, implement, and improve CTE;
- conducting and disseminating national research and disseminating information on best practices that improve CTE programs and programs of study, services, and activities;
- providing technical assistance that promotes leadership, initial preparation, and professional development at the state and local levels, and improves the quality of CTE teachers, faculty, administrators, and counselors;
- supporting partnerships among secondary schools, postsecondary institutions, baccalaureate degree granting institutions, area CTE schools, local workforce investment boards,[16] business and industry, and intermediaries;
- providing individuals with opportunities throughout their lifetimes to develop, in conjunction with other education and training programs, the knowledge and skills needed to keep the United States competitive; and
- increasing the employment opportunities for populations who are chronically unemployed or underemployed, including individuals with disabilities, individuals from economically disadvantaged families, out-of-workforce individuals,[17] youth who are in, or have aged out of, the foster care system, and homeless individuals.

[12] Perkins V, §113.

[13] Perkins V, §116

[14] Student participation in CTE programs is also supported by the federal government at the postsecondary level through federal student aid.

[15] Perkins IV, §2, and Perkins V, §6.

[16] Workforce investment boards (WIBs) were established under the Workforce Investment Act of 1998 (WIA, P.L. 105-220) to administer employment and training services under the act. The Workforce Innovation and Opportunity Act (WIOA, P.L. 113-128) succeeded WIA as the primary federal workforce development statute. Under the WIOA, entities called workforce development boards (WDBs) perform functions similar to WIBs under WIA.

[17] An *out-of-workforce individual* means a displaced homemaker, as defined in Section 3 of the WIOA; "or an individual who has worked primarily without remuneration to care for a home and family, and for that reason has diminished marketable skills; or is a parent whose youngest dependent child will become ineligible to receive assistance under part A of title IV of the Social Security Act (42 U.S.C. 601 et seq.) not later than 2 years after the date (continued...)

The following sections provide a detailed description of the Title I Basic State Grants program, discussion of other Title I programs, and information about additional provisions, authorizations, and appropriations levels.

Title I—Basic State Grants Program

The BSG program is a formula grant program that receives over 90% of the aggregate funds appropriated for Perkins V. The uses of BSG funds are closely aligned with the stated purpose of Perkins V discussed above.

This section of the report describes the BSG program funding allocation formulas, planning activities at the state and local levels, uses of funds, and reporting and accountability requirements.

Brief Overview of Allocation Procedures

After set-asides for some of the outlying areas and Native American and Native Hawaiian programs, BSG grant funds are allocated to the 50 states, the District of Columbia (DC), Puerto Rico, and the U.S. Virgin Islands. Funds are allocated initially through foundational grants equal to FY2018 grant allocations (ratably reduced if insufficient funding is available). When appropriations are in excess of the amount of funding needed to provide foundational grants, the excess appropriations (hereinafter referred to as *new money*) are then, subject to some small state minimum grant requirements, allocated according to population and per capita income through a factor-based formula discussed below.

Reservations

Set-Aside for Guam, American Samoa, the Northern Mariana Islands, and Palau[18]

Outlying areas in Section 3(21) refer to the U. S. Virgin Islands, Guam, American Samoa, the Commonwealth of the Northern Mariana Islands, and the Republic of Palau. However, the U.S. Virgin Islands are defined as a state for the purposes of the BSG formula (Perkins V, §111(d)) and receive funds under the state grant formula.

The total set-aside for Guam, American Samoa, the Commonwealth of the Northern Mariana Islands, and the Republic of Palau is 0.13% of the amount appropriated for BSG. From this set-aside, Guam receives an initial allotment of $660,000; American Samoa and the Northern Mariana Islands receive $350,000 each; and the Republic of Palau receives $160,000. The remainder of the set-aside is divided equally among the aforementioned outlying areas.

Set-Aside for Native American and Native Hawaiian Programs[19]

The Secretary of Education (hereinafter referred to as the Secretary) is required to reserve 1.25% of BSG funds for the Native American Career and Technical Education Program (NACTEP). Eligible entities for NACTEP funds include federally recognized Indian tribes, tribal

on which the parent applies for assistance under such title; and is unemployed or underemployed and is experiencing difficulty in obtaining or upgrading employment."

[18] Perkins V, §114.

[19] Perkins V, §115.

organizations, Alaska Native entities, BIE schools,[20] and consortia of such entities. The funds are awarded competitively. NACTEP funds may be used to carry out CTE programs and services or to provide direct assistance to CTE students.[21]

The Native Hawaiian CTE Program (NHCTEP), which receives a 0.25% reservation from the total BSG appropriation, requires the Secretary to award grants to community-based organizations serving and representing Native Hawaiians to plan and administer CTE programs that benefit Native Hawaiians.[22] The funds are awarded competitively.

State Allocation Formula[23]

After reservations for Guam, American Samoa, the Commonwealth of the Northern Mariana Islands, the Republic of Palau, and the Native American and Native Hawaiian CTE Programs, BSG funds are awarded to the 50 states, DC, Puerto Rico, and the U.S. Virgin Islands. Each of the 50 states, DC, Puerto Rico, and the U.S. Virgin Islands receives a foundational grant equal to its FY2018 BSG allocation. If the appropriation is less than the FY2018 appropriation, such foundational grant allocations are ratably reduced.

If appropriations exceed the FY2018 appropriations level, up to one-third of the new money (i.e., appropriations in excess of the FY2018 BSG appropriations level) would first be allocated to states that received less than 0.5% of total appropriations. Then, either when all states would have reached the 0.5% level, or when one-third of the new money would be allocated, at least two-thirds of the new money plus any remainder would be allocated based on states' population and per capita income (PCI) factors. The formula awards proportionally larger amounts to states with larger populations in the age range traditionally targeted by CTE programs (15 to 19 years old) and to states with lower than average PCIs.

See **Appendix B** for a more detailed explanation of this formula, and **Appendix C** for the actual FY2020 BSG state allocations.

Allocation of State and Local Funds

Perkins V requires states to allocate

- at least 85% to the local level,
- up to 10% for state leadership activities,[24] of which up to 2% can serve individuals in state correctional institutions, juvenile justice facilities, and institutions that serve students with disabilities; and
- up to 5% or $250,000 (whichever is greater) for program administration.

[20] Additionally, schools that are funded by the BIE are also eligible for NACTEP funds. However, they cannot use these funds to carry out secondary-level CTE programs, because those programs already receive money through the states.

[21] For NACTEP grantees, see http://cte.ed.gov/grants/native-american-career-and-technical-education-program.

[22] For a list of grantees, see http://cte.ed.gov/grants/native-hawaiian-career-and-technical-education-program.

[23] Perkins V, §111.

[24] These funds can be used for several purposes, including supporting preparation for non-traditional fields, developing programs for special populations, recruiting CTE faculty and staff, and establishing statewide articulation agreements and industry partnerships. See Perkins V, Section 124 for a complete list of permissible uses; also see the "State Leadership Activities" subsection in this report.

Within-State Distribution[25]

States decide the split of local level funds between CTE providers at the secondary and postsecondary levels. These proportions vary widely from state to state.[26]

Additionally, up to 15% of the funds available for allocation at the local level may be awarded to support CTE in a collection of specified settings. These funds may be awarded to local CTE providers that are in rural areas, or providers in areas with high percentages of CTE students, areas with high numbers of CTE students, and areas with disparities or gaps in performance. The statute does not specify how the states are to allocate these funds to eligible recipients.

Distribution at the Secondary Education Level

Eligible recipients of funds at the secondary education level are local educational agencies (LEAs), including public charter schools that operate as LEAs; BIE schools; area CTE schools providing secondary education;[27] educational service agencies;[28] or consortia of such entities.

The formula for distributing money to the secondary level[29] is

- 70% of the funds are allocated based on each local recipient's share of individuals aged 5 through 17 living below the poverty line, compared to the total population of individuals aged 5 through 17 living below the poverty line in the entire state; and
- 30% of the funds are allocated based on each local recipient's share of all individuals aged 5 through 17.[30]

If a secondary level local recipient's calculated allotment is less than $15,000, the recipient must form a consortium with other eligible secondary level recipients in order to meet the minimum grant requirement. This requirement can be waived if the recipient is unable to enter into a consortium and is either located in a rural, sparsely populated area or is a public charter school operating a CTE program.

Distribution at the Postsecondary Education Level

Eligible recipients of funds at the postsecondary education level are public or nonprofit institutions of higher education (IHEs), LEAs providing postsecondary education, area CTE schools providing postsecondary education, tribally controlled colleges and universities, educational service agencies, and consortia of such entities.

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[25] Perkins V, §111.

[26] In academic year 2019-2020, 13 states allocated 50% or more to postsecondary education. The largest postsecondary level allocation share was Colorado's (60%), while the smallest was Rhode Island's (10%). (Perkins Collaborative Resource Network data https://cte.ed.gov/grants/state-allocations as analyzed by CRS.)

[27] Secondary area CTE schools are either specialized public secondary schools used exclusively or principally for the provision of CTE to individuals who are available for study in preparation for entering the labor market, or the department of a public secondary school exclusively or principally used for providing CTE in not fewer than three different fields that are available to all students, especially in high-skill, high-wage, or in-demand industry sectors or occupations.

[28] An educational service agency is a regional public multiservice agency authorized by state statute to develop, manage, and provide services or programs to LEAs.

[29] Perkins V, §131.

[30] The Secretary may permit a state to use an alternative secondary distribution formula if that formula "more effectively targets funds on the basis of poverty" (Perkins IV, §131(b)(1)).

Funds are distributed to eligible local recipients at the postsecondary level based on their share of Pell Grants recipients and recipients of assistance from the BIE who are enrolled in CTE programs.[31] The minimum grant amount is $50,000.

If a state chooses to distribute 15% or less of its total allocation to the postsecondary level, that state may allocate funds on a competitive basis or using any alternative method.

State Planning and State Leadership Activities

This section describes required components of state plans as well as requirements and permissible uses of funds for state leadership activities.

State Plans[32]

In order to receive funding, each eligible agency[33] must submit a four-year state plan to the Secretary. These plans vary widely in composition, scope, and allocation of funding between secondary and postsecondary CTE, including whether they have statewide articulation agreements.[34]All plans must be developed through public hearings and in consultation with the governor and appropriate state agencies, as well as a variety of stakeholders, including CTE teachers, faculty, and administrators; career guidance and academic counselors; eligible recipients;[35] parents and students; representatives of special populations;[36] representatives of business, industry, and labor organizations; representatives of two-year minority-serving IHEs and, where possible, historically Black and tribally controlled colleges and universities; representatives of the state workforce development board; representatives of agencies serving out-of-school youth, homeless children, and students with disabilities; and representatives of Indian tribes and tribal organizations that may be present in the state.[37] The plan contents must, among other requirements, include such information as a description of

- supported CTE activities, including the degree to which programs are aligned with and address the education and skill needs of the employers identified by the state board;
- a strategic vision and set of goals for preparing an educated and skilled workforce (including special populations) and for meeting the skilled workforce

[31] For more information, see CRS Report R45418, *Federal Pell Grant Program of the Higher Education Act: Primer.*

[32] Perkins V, §122.

[33] An *eligible agency* is defined in Section 3(12) of Perkins IV as "a State board designated or created consistent with State law as the sole State agency responsible for the administration of career and technical education in the State or for the supervision of the administration of career and technical education in the State."

[34] Components of different state plans can be compared at https://cte.ed.gov/dataexplorer/build using the Perkins State Plans and Data Explorer website.

[35] Eligible recipients are an LEA (including a public charter school that operates as an LEA), an area CTE school, an educational service agency, an Indian tribe, tribal organization, or tribal educational agency or a consortium, eligible to receive assistance under Section 131, or an eligible institution or consortium of eligible institutions eligible to receive assistance under Section 132.

[36] The term *special populations* is defined in Section 3(29) of Perkins IV and means individuals with disabilities; individuals from economically disadvantaged families, including low-income youth and adults; individuals preparing for non-traditional fields; single parents, including pregnant women; out-of-workforce individuals; English learners; homeless individuals; youth who are in, or have aged out of, the foster care system; and youth with a parent who is a member of the Armed Forces or is on active duty.

[37] For more information on how this stakeholder definition has expanded since Perkins IV, see CRS Report R45446, *Reauthorization of the Perkins Act in the 115th Congress: The Strengthening Career and Technical Education for the 21st Century Act*.

needs of employers, including in existing and emerging in-demand industry sectors and occupations as identified by the state;

- a strategy for any joint planning, alignment, and coordination of funds between CTE programs with the state's workforce development system and other federal programs;
- CTE programs of study to be offered by the state (for more information on programs of study, see the text box below);
- professional development, recruitment, and retention strategies for CTE teachers, faculty, administrators, and career and guidance counselors;
- efforts to facilitate the successful transition of CTE students from sub-baccalaureate programs into baccalaureate degree programs at IHEs;
- how the state will annually evaluate the effectiveness of its CTE programs; and
- the state's CTE programs aimed at members of special populations.

Programs of Study

According to Section 7(41) of Perkins V, programs of study (POS) are coordinated, nonduplicative sequences of academic and technical content at the secondary and postsecondary level that

- incorporate challenging academic standards;
- address both academic and technical knowledge and skills, including employability skills;
- are aligned with the needs of industries;
- progress in specificity;
- have multiple entry and exit points that incorporate credentialing; and
- culminate in the attainment of a recognized postsecondary credential.

POS are not federally prescribed curricula.[38] A POS may include the opportunity for secondary education students to acquire postsecondary credits through dual or concurrent enrollment programs or in other ways.

Eligible CTE providers at the local level must offer the elements of at least one POS that is described by the state in its state plan.

As an alternative to the state plan described above, a state may submit a state plan as part of the combined plan submitted under Section 103 of the Workforce Innovation and Opportunity Act (WIOA; P.L. 113-128).[39]

The state plan is approved by the Secretary of Education unless it does not meet the minimum requirements of Perkins V related to core performance indicators.[40] The Secretary may disapprove a state plan only after giving the state agency notice and an opportunity for a hearing. At the end of the period covered by the first four-year plan, a state may submit a subsequent four-year plan

[38] Section 438 of the General Education Provisions Act (20 U.S.C. §1232a) explicitly states that no provision included in the majority of federal education programs administered by the U.S. Department of Education is intended to authorize the federal government to exercise control over curriculum, instructional programs, administration, or school personnel or in the selection of printed material, or in the assignment of transportation of students or teachers to overcome a racial imbalance.

[39] Nine states chose to submit a combined plan in FY2020: Alabama, Delaware, Indiana, Minnesota, Ohio, Pennsylvania, Rhode Island, Virginia, and Washington. The WIOA is the primary federal workforce development law intended to bring about increased coordination among federal workforce development and related programs. The section of the combined state plan that relates to CTE would still have to follow the same requirements as a single plan. For more information on the WIOA, see CRS Report R44252, *The Workforce Innovation and Opportunity Act and the One-Stop Delivery System*.

[40] Perkins V, §122. See also the "Core Indicators of Performance" section of this report for more detail on the few conditions under which the Secretary may disapprove a state plan.

or submit for secretarial approval annual revisions to only the state-determined levels of performance.

State Leadership Activities[41]

States may reserve up to 10% of their allotments for state leadership activities. Of these funds, an amount equal to not more than 2% of the total state allotment must be used for individuals in state institutions, such as state correctional institutions, juvenile justice facilities, and educational institutions serving individuals with disabilities. States are required to reserve at least the lesser of 0.1% of their state allocations or $50,000 for recruiting members of special populations to enroll in CTE programs. Additionally, between $60,000 and $150,000 must be reserved for services preparing individuals for nontraditional fields.[42]

State leadership activities must include support for

- "(A) preparation for non-traditional fields in current and emerging professions, programs for special populations, and other activities that expose students, including special populations, to high-skill, high-wage, and in-demand occupations;
- (B) individuals in State institutions, such as State correctional institutions, including juvenile justice facilities, and educational institutions that serve individuals with disabilities;
- (C) recruiting, preparing, or retaining career and technical education teachers, faculty, specialized instructional support personnel, or paraprofessionals, such as preservice, professional development, or leadership development programs;
- (D) technical assistance for eligible recipients;" and
- a report on the effectiveness of such use of funds in achieving the goals described in the state's strategic vision as described in the state plan and the state determined levels of performance, and reducing disparities or performance gaps between student subgroups[43] (for more information, see the "Accountability" section of this report).

In addition to these required activities, there are several permissible state leadership activities outlined in Section 124. These include

- developing statewide programs of study, which may include standards, curriculum, and course development, and career exploration, guidance, and advisement activities and resources;
- establishing statewide articulation agreements aligned to approved, locally developed programs of study;
- establishing statewide industry or sector partnerships among LEAs, IHEs, adult education providers, Indian tribes and tribal organizations, employers, and parents;

[41] Perkins V, §124.

[42] Nontraditional fields are occupations or fields of work, such as careers in computer science, technology, and other current and emerging high-skill occupations, for which individuals from one gender comprise less than 25% of the individuals employed in each such occupation or field of work.

[43] Perkins V, §124(1).

- high-quality comprehensive professional development for teachers and staff that is, to the extent practicable, grounded in evidence-based research;
- awarding incentive grants to eligible recipients;
- "the creation, implementation, and support of pay for success initiatives leading to a recognized postsecondary credential";
- "support for programs and activities that increase access, student engagement, and success in science, technology, engineering, and mathematics fields (including computer science, coding, and architecture), support for the integration of arts and design skills, and support for hands-on learning, particularly for students who are members of groups underrepresented in such subject fields, such as female students, minority students, and students who are members of special populations";
- "making all forms of instructional content widely available, which may include use of open educational resources;
- developing valid and reliable assessments of competencies and technical skills and enhancing data systems to collect and analyze data on secondary and postsecondary academic and employment outcomes;
- support for accelerated learning programs";
- support for career academies; and
- "other state leadership activities that improve CTE."

Local Activities

Local Plan[44]

In order to receive Perkins funding, an eligible recipient at the local level must submit a local plan to the state eligible agency. Plans must be developed in consultation with a diverse group of stakeholders. The local plan contents must include such information as a description of

- the results of the comprehensive needs assessment[45];
- "information on the career and technical education course offerings and activities that the eligible recipient will provide with funds under this part, which shall include not less than 1 program of study approved by a state,"
- collaboration with local workforce development boards and other local workforce agencies, one-stop delivery systems,[46] and other partners, to provide

[44] Perkins V, §134.

[45] A comprehensive needs assessment must include an evaluation of the performance of the students served with respect to state and local levels of performance, which are the quantifiable indicators decided upon by the state and either accepted or negotiated by each locality. These levels of performance include goals for high school graduation, enrollment in postsecondary education, earned credentials, and other outcomes discussed in the "Core Indicators of Performance" section. The needs assessment must include an evaluation for special populations and each subgroup described in the Elementary and Secondary Education Act (ESEA); a description of how the eligible recipient will improve recruitment, retention, and training of CTE faculty and staff; and a description of progress toward the implementation of CTE programs and programs of study. Subgroups under the ESEA are each major and racial ethnic group, economically disadvantaged students, students with disabilities, and students with limited English proficiency.

[46] For more information, see CRS Report R44252, *The Workforce Innovation and Opportunity Act and the One-Stop Delivery System*.

    - "career exploration and career development coursework, activities, or services;
    - career information on employment opportunities that incorporate the most up-to-date information on high-skill, high-wage, or in-demand industry sectors or occupations, as determined by the comprehensive needs assessment"; and
    - "an organized system of career guidance and academic counseling to students before enrolling and while participating in a career and technical education program";
- how the eligible recipient will improve the academic and technical skills of CTE students through the integration of standards-aligned content to ensure learning in the subjects that constitute a well-rounded education[47];
- how the eligible recipient will provide activities to prepare special populations for high-skill, high-wage, or in-demand industry sectors or occupations that will lead to self-sufficiency, prepare CTE participants for non-traditional fields, provide equal access for special populations to CTE, and ensure that members of special populations will not be discriminated against on the basis of their status as members of special populations;
- work-based learning opportunities and opportunities to gain postsecondary credit while in high school, such as through dual or concurrent enrollment programs or early college high school;
- "how the eligible recipient will coordinate with the eligible agency and institutions of higher education to support the recruitment, preparation, retention, and training, including professional development, of teachers, faculty, administrators, and specialized instructional support personnel and paraprofessionals who meet applicable state certification and licensure requirements (including any requirements met through alternative routes to certification), including individuals from groups underrepresented in the teaching profession"; and
- how the eligible recipient will address disparities or gaps in performance as described in each of the plan years.

Local Uses of Funds[48]

As at the state level, eligible recipients (also known as local providers) at the local level have a list of required uses of funds. They must develop, coordinate, implement, or improve CTE programs that meet the needs identified in the comprehensive needs assessment and that are of sufficient size, scope, and quality to be effective. These programs must

- "provide career exploration and career development activities through an organized, systematic framework designed to aid students, including in the middle grades, before enrolling and while participating in a career and technical education program, in making informed plans and decisions about future education and career opportunities and programs of study";

[47] As defined in Section 8101 of the ESEA.

[48] Perkins V, §135.

- "provide professional development for teachers, faculty, school leaders, administrators, specialized instructional support personnel, career guidance and academic counselors, or paraprofessionals";
- provide CTE students the skills necessary to pursue careers in high-skill, high-wage, or in-demand industry sectors or occupations;
- support the integration of academic skills into CTE programs and programs of study;
- support the implementation of CTE programs and programs of study that increase local levels of performance; and
- develop and implement evaluations, including evaluations necessary to complete the comprehensive needs assessment.

Each eligible recipient may use up to 5% of its local funds for administrative costs.

Accountability

Under Perkins V, states and local CTE providers that receive Perkins funds are required to satisfy certain accountability requirements based on student outcomes for CTE concentrators. The accountability requirements in Section 112 are intended to optimize the return on investment of federal CTE funds.

While Perkins V mandates the inclusion of core indicators, each state decides on the appropriate levels of performance for each of the core indicators described below, as well as any additional indicators included in its approved plan. The Secretary of Education cannot assist in establishing these levels unless requested by the state, but the levels should take into account how they compare with the levels in other states.

For example, a state may decide upon a 70% statewide goal for CTE concentrators to earn an industry-recognized postsecondary credential. **Table 1** provides one such example. Localities must either accept these state-determined levels of performance or negotiate with the state to reach agreement on new levels of performance.

Table 1. Sample State-Determined Levels of Postsecondary Performance

Indicator	FY2021	FY2022	FY2023	FY2024
Postsecondary Retention and Placement	69.0%	69.4%	69.8%	70.4%
Earned Industry-Recognized Postsecondary Credential	70.1%	70.5%	70.9%	71.3%
Nontraditional Program Enrollment	9.6%	9.8%	9.9%	10.1%

Source: Illinois Community College Board, at http://www2.iccb.org/cte/perkins/data-and-accountability/.

Note: Each of these indicators refers to CTE concentrators, meaning those who either earned 12 CTE credits or completed a CTE degree or certificate requiring fewer than 12 credits.

Both the state and the local provider are held accountable for meeting at least 90% of the level of performance for each indicator. If a state or local provider fails to meet at least 90% of the levels of performance for one or more indicators, the state or local provider is required to develop and

implement an improvement plan. Funds may be withheld from a state or local provider that subsequently fails to meet certain criteria.

Core Indicators of Performance[49]

The core indicators of performance at the secondary level are

- the four-year adjusted cohort high school graduation rate for CTE concentrators;[50]
- the proficiency of CTE concentrators as measured by state academic performance standards on state tests in accordance with Title I of the Elementary and Secondary Education Act (ESEA);
- "the percentage of CTE concentrators who, in the second quarter after exiting from secondary education, are in postsecondary education or advanced training, military service or a service program that receives assistance under title I of the National and Community Service Act of 1990 (42 U.S.C. 12511 et seq.), are volunteers as described in section 5(a) of the Peace Corps Act (22 U.S.C. 2504(a)), or are employed";
- "the percentage of CTE concentrators in CTE programs or programs of study that lead to non-traditional fields"; and
- at least one of the following:
  - "[t]he percentage of CTE concentrators graduating from high school having attained a recognized postsecondary credential,"
  - the percentage of CTE concentrators graduating from high school having attained postsecondary credits through a dual or concurrent enrollment program or another credit transfer agreement, or
  - "The percentage of CTE concentrators graduating from high school having participated in work-based learning."

The core indicators of performance at the postsecondary level are

- "(i) The percentage of CTE concentrators who, during the second quarter after program completion, remain enrolled in postsecondary education, are in advanced training, military service, or a service program that receives assistance under title I of the National and Community Service Act of 1990 (42 U.S.C. 12511 et seq.), are volunteers as described in section 5(a) of the Peace Corps Act (22 U.S.C. 2504(a)), or are placed or retained in employment.
- (ii) The percentage of CTE concentrators who receive a recognized postsecondary credential during participation in a program or within 1 year of program completion.
- (iii) the percentage of CTE concentrators in career and technical education programs and programs of study that lead to non-traditional fields."

[49] Perkins V, §112. A list is also available at https://cte.ed.gov/accountability/core-indicators.

[50] The four-year adjusted cohort high school graduation rate is as defined in Section 8101 of the ESEA as the number of students who graduate in four years with a regular high school diploma divided by the number of students who form the adjusted cohort for the graduating class. In addition to using the four-year adjusted cohort graduation rate, a state may also choose to use a graduation rate based on a period longer than four years as a core indicator of performance.

In addition to these required indicators, states may choose to include additional indicators of performance in their state plans. Through the state plan development process, each eligible agency establishes levels of performance (goals) for each of the core indicators of performance. These levels of performance must

- "be objective, quantifiable, and measurable";
- require the state to continually make meaningful progress toward improving the performance of all CTE students, including subgroups and special populations; and
- have been subject to public comment, to which the eligible agency has provided a written response.

The Secretary shall approve state plans no later than 120 days after submission and can only disapprove a plan if the state-determined levels of performance fail to meet the requirements listed above. After initial approval, states may submit adjustments to their levels of performance prior to the third year. Adjusted levels of performance must meet the above requirements of the initial levels of performance and take into account

- how the levels of performance involved compare with the levels of performance established for other states,
- factors including the characteristics of actual (as opposed to anticipated) CTE concentrators; and
- the services to be provided.

The adjusted levels must be higher than the average actual performance of the two most recently completed program years, except in the case of unanticipated circumstances.[51] The Secretary shall approve the revised levels of performance if those levels meet the requirements.

When establishing local levels of performance, each eligible recipient agrees to accept the state determined levels of performance for each year of the plan, or negotiate with the state to reach agreement on new local levels of performance, for each of the core indicators of performance. Eligible recipients may, with agreement from the eligible agency, adjust the local levels of performance prior to the third year of the local application.

Data Reporting and Disaggregation

States annually report to the Secretary on the levels of performance achieved by their CTE concentrators on each of the core indicators of performance, as well as any additional indicators of performance selected by the state as part of its submitted plan. Each level of performance must be disaggregated by special populations, CTE program, student level (e.g., secondary), award level (e.g., associate's degree), ESEA Title I subgroups, and other categories.[52] The data in the annual state reports, known as Consolidated Annual Reports, must be publicly disseminated by the Secretary. State performance data must also be included in an annual report to Congress produced by the Secretary in accordance with Section 112 of Perkins V.

Localities submit similar reports to their states. These reports must be made available by the eligible recipient through a variety of formats, including online, to students, parents, educators,

[51] *Unanticipated circumstances* are not defined in Perkins V.

[52] These include each major racial and ethnic group, economically disadvantaged students, children with disabilities, and English learners. Performance data must also be disaggregated by gender and migrant status.

and the public, in a format that is uniform and, to the extent practicable, provided in a language that students, parents, and educators can understand.

Sanctions

States that do not meet 90% of a state level of performance for any of the core indicators of performance in a given year are required to develop and implement a program improvement plan during the following program year. The plan is to be developed with input from various stakeholders. The Secretary must provide technical assistance to the state if it is determined that the state is not making substantial progress. The Secretary may withhold all, or a portion of, a state's leadership and administrative funds if either of the following occurs:

- the state fails to implement a program improvement plan; or
- the state fails to meet at least 90% of a level of performance for the same core indicator of performance for two consecutive years after being identified for improvement.

The Secretary must use the withheld funds to provide technical assistance or other improvement assistance to the state. The Secretary may waive these sanctions due to exceptional or uncontrollable circumstances, such as a natural disaster or a precipitous and unforeseen decline in the financial resources of the state.[53]

Similarly, a local provider that does not meet at least 90% of its performance levels on at least one core indicator of performance must also develop a program improvement plan. After providing technical assistance, and if the eligible recipient fails to implement the improvement plan or fails to meet at least 90% of the specified level of performance for two consecutive years following identification for improvement, the state may withhold some or all of the local provider's funding and use it to provide CTE services for the affected students through alternative means. The state may also waive these sanctions due to exceptional or uncontrollable circumstances.

National Activities and Other Title I Programs

National Activities[54]

Perkins V authorizes the Secretary through the U.S. Department of Education's (ED's) Director of the Institute for Education Sciences (IES) to carry out the following national activities:

- create a single plan for the research, development, dissemination, evaluation, and assessment of CTE programs in consultation with, or acting through, ED's Director of the IES;
- provide an annual report to Congress and other stakeholders on the state of CTE and on state and local performance of Perkins-funded CTE programs, including a quantitative analysis of the effectiveness of activities funded by Perkins V;[55]

[53] Perkins V, §201.

[54] Perkins V, §113.

[55] The Perkins reports to Congress are available at https://cte.ed.gov/accountability/reports-to-congress.

- direct ED's National Center for Education Statistics (NCES) to collect information on CTE for a nationally representative sample of students as part of its regular assessments, which may include international comparisons;[56]
- conduct an evaluation that describes
  - the extent and success of the integration of state academic standards, including a review of academic and technical proficiency achievement;
  - the number of graduates (high school diplomas and postsecondary credentials) and CTE participants;
  - the extent to which CTE programs prepare students for employment or postsecondary education;
  - employer involvement and satisfaction with CTE programs;
  - efforts to expand access to CTE programs to all students;
  - innovative approaches to work-based learning that increase participation and alignment with employment in high-growth industries, including in rural and low-income areas;
  - the effectiveness of different delivery systems and approaches (e.g., dual enrollment or career academies);
  - the extent to which CTE programs are grounded in evidence-based research;
  - the impact of Perkins V compared to Perkins IV, including comparisons of the use of comprehensive needs assessments, implementation of programs of study, and coordination of planning and program delivery with other relevant laws, including the WIOA and the ESEA;
  - changes in CTE program accountability and the effect of such changes on program delivery and quality;
  - changes in student enrollment patterns; and
  - efforts to reduce disparities or performance gaps between subgroups, special populations, and other disaggregated groups.

The Secretary must also appoint an independent advisory panel to conduct an analysis of the findings and recommendations resulting from the evaluation described above.

The Secretary may collect and disseminate information on states' efforts to meet their determined levels of performance. After consultation with the IES Director, the Commissioner for Education Research, and the states, and with input from the independent advisory panel, the Secretary shall award a grant, contract, or cooperative agreement, on a competitive basis, to an IHE or a consortium of one or more IHEs and one or more private nonprofit organizations or agencies, to carry out evidence-based research.[57]

The Secretary, in consultation with the IES Director, was to submit to the relevant committees of Congress an interim report regarding the evaluation and summary of research activities carried

[56] The website summarizing CTE statistics is available at http://nces.ed.gov/surveys/ctes/.

[57] See the research and efforts of the National Research Center for Career and Technical Education at https://www.sreb.org/national-research-center-career-and-technical-education. Examples of evidence-based CTE research include using longitudinal data systems to understand whether CTE programs produce better outcomes in terms of enrollment, employment, and wages. This research may also help to develop, improve, and identify the most successful methods for eliminating inequities in access to effective teaching in CTE programs, and for addressing the needs of special populations.

out under this section no later than two years after the date of enactment of Perkins V. No later than four years after the date of enactment of Perkins V, the Secretary shall submit a final report summarizing the studies and analyses that relate to the evaluation and summary of research activities carried out under Section 113.[58] This report must be updated biennially.

Innovation and Modernization Grants

Though the vast majority of Perkins V funding is awarded on a noncompetitive basis via BSGs and approved state plans, Innovation and Modernization (I&M) grants represent a different approach designed to encourage experimentation, as these federal grants can be administered directly to LEAs or IHEs. The purpose of the competitive I&M grants[59] is to identify, support, and rigorously evaluate evidence-based and innovative strategies and activities to improve and modernize CTE and align workforce skills with labor market needs. No more than 20% of the national activities funding may be used for I&M grants.

Eligible entities, institutions, and recipients are as defined throughout Perkins V, and previous recipients of these competitive grants include LEAs, IHEs, and state boards. Applicants that predominately serve low-income families have priority in the I&M grant competition. The first competition occurred in 2019, with nine grants awarded, each totaling $1.5 million; all grantees received approximately $500,000 over the course of three years.[60] Grant activities must be independently evaluated, including a quantitative analysis of the effectiveness of the project carried out under the I&M grant.

Grantees must match not less than 50% of the federal funds with nonfederal sources, and they must show how the CTE programs or programs of study to be implemented reflect the needs of employers, as demonstrated by a comprehensive needs assessment. Funds must be used to create, develop, implement, replicate, or take to scale evidence-based, field-initiated innovations to modernize and improve the effectiveness and alignment of CTE education and to improve student outcomes. These efforts may include the following activities:[61]

- designing and implementing courses or programs of study aligned to labor market needs in new or emerging fields and working with industry representatives to upgrade equipment, technology, and related curricula used in CTE programs, with an emphasis on coding and computer science subject areas;
- improving CTE outcomes through innovative delivery models, including school-based simulated work sites, mentoring, work site visits, job shadowing, project-based learning, and skills-based and paid internships;
- improving the transition of students from secondary to postsecondary education or employment, or from the completion of one postsecondary program to another, which could include the creation, development, or expansion of dual or concurrent enrollment programs;
- redesigning the high school experience, which may include evidence-based transitional support strategies for students who have not met postsecondary eligibility requirements;

[58] For an example, see the final report authorized under and published consistent with Perkins IV at http://s3.amazonaws.com/PCRN/docs/NACTE_FinalReport2014.pdf.

[59] I&M grants did not exist prior to Perkins V.

[60] For a list of grantees and more information on their applications, see https://cte.ed.gov/grants/innovation-and-modernization-grant-program.

[61] A comprehensive list of allowable grantee activities can be found in Perkins V, Section 113.

- creating or expanding recruitment, retention, or professional development activities for CTE faculty and support staff, including counselors and paraprofessionals;
- developing a coordinated strategy that uses CTE programs to address identified employer needs and workforce shortages, such as shortages in the early childhood, elementary school, and secondary school education workforce; and
- establishing an online portal for CTE students, including special populations, to prepare them for postsecondary CTE education, which could include mentoring, identifying career opportunities and interests, and a platform to establish online savings accounts and gain financial literacy skills.

Tribally Controlled Postsecondary Career and Technical Institutions Program (TCPCTIP)[62]

Section 116 of Perkins V authorizes grants to provide support for the education and training of Indian students attending tribally controlled postsecondary career and technical institutions that are not eligible for assistance under Title I of the Tribally Controlled Colleges and Universities Assistance Act of 1978 (TCCUAA)[63] or the Navajo Community College Act.[64] There are two IHEs that qualify for funding under TCPCTIP: the United Tribes Technical College in North Dakota and the Navajo Technical University in New Mexico. These two IHEs are also the only recipients of funds from the Department of the Interior's Tribal Technical Colleges program.[65]

Funds provided under TCPCTIP may be used by recipients for expenses associated with CTE programs for Indian students, including program development costs; operations and maintenance of facilities; equipment, transportation, daycare, and family support programs for students; and student stipends.

Additional Provisions under Title II

Title II of Perkins V specifies general provisions regarding federal and state administrative provisions. The following subsections describe a few key provisions.

Supplement, Not Supplant and Maintenance of Effort[66]

The supplement, not supplant provision requires states and local education providers to supplement, not supplant, nonfederal funds to carry out CTE activities.

The maintenance of effort (MOE) provision states that to receive Perkins funds, a state's fiscal effort per CTE student or aggregate CTE expenditures for the preceding fiscal year must be equal to or greater than the corresponding expenditures from the preceding fiscal year. The state may establish a new baseline for the MOE metrics, which is not less than 95% of the state's fiscal effort per CTE student or aggregate CTE expenditures for the preceding fiscal year, or the state may use the baseline that was in effect on the day before the enactment of Perkins V. If total

[62] Perkins V, §116.

[63] 25 U.S.C. §1801 et seq.

[64] P.L. 92-189, as amended.

[65] TCCUAA, Title V; 25 U.S.C. §1861 et seq.

[66] Perkins IV, §311.

Perkins V CTE program appropriations decrease by a certain percentage, states may decrease their fiscal effort per CTE student or aggregate CTE expenditures by the same percentage.

If a state fails to meet its MOE requirement for one or more of the five preceding fiscal years, the state's allotment is reduced by the same proportion by which the state failed to meet the requirement. The Secretary can waive this provision under exceptional circumstances.

Limitation for Certain Students[67]

Perkins V funds may not be used to provide CTE programs for students prior to the middle grades (grades 5-8, as defined in Section 8101 of the ESEA).

Private Schools Staff and Students[68]

State and local CTE providers are to include, "to the extent practicable upon written request," teachers and other personnel at nonprofit private schools in local CTE professional development activities. Similarly, this section allows for the "meaningful participation" of students attending nonprofit private schools in local CTE programs funded by Perkins V.

Study on Programs of Study Aligned to High-Skill, High-Wage Occupations[69]

The Government Accountability Office (GAO) must conduct a study to evaluate the strategies, components, policies, and practices used by eligible agencies or eligible recipients receiving Perkins funding to successfully assist all students in pursuing and completing programs of study aligned to high-skill, high-wage occupations, including subgroups or special populations who may be underrepresented in these fields. This study must also note any challenges associated with the replication of best practices and must be informed by geographically diverse and representative groups of students, parents, Perkins recipients, CTE teachers and support staff, Indian tribes and tribal organizations, special populations, and representatives of business and industry. The Comptroller General must submit the study to the House Committee on Education and the Workforce and to the Senate Committee on Health, Education, Labor, and Pensions.[70]

Perkins V Authorizations and Appropriations

Perkins V contains three separate authorizations of appropriations: Basic State Grants, National Activities, and Tribally Controlled Postsecondary Career and Technical Institutions (TCPCTI). Appropriations are authorized for FY2019 through FY2024.

Table 2 includes actual appropriations for FY2019-FY2021 and budget authority for FY2022-FY2024.

[67] Perkins V, Section 215.

[68] Perkins IV, §317.

[69] Perkins V, §219.

[70] There is no statutorily specified deadline for when GAO must submit the results of the study.

Table 2. Perkins V Actual Appropriations and Authorizations of Appropriations, FY2019-FY2024

(Dollars in thousands)

Program	FY2019 (Actual)	FY2020 (Actual)	FY2021 (Actual)	FY2022 (Authorized)	FY2023 (Authorized)	FY2024 (Authorized)
Basic State Grants (Title I)	1,262,598	1,282,598	1,334,848	1,281,937	1,299,884	1,318,082
National Activities (§113)	7,421	7,421	7,421	7,977	8,089	8,202
TCPCTI	9,564	10,000	10,634	10,178	10,321	10,456
Total, Perkins IV	**1,279,583**	**1,300,019**	**1,352,903**	**1,300,092**	**1,318,294**	**1,336,740**

Sources: U.S. Department of Education, Department of Education Budget Tables, *FY 2021 Congressional Action*, https://www2.ed.gov/about/overview/budget/tables.html?src=rt, and Perkins V (P.L. 115-224)

Note: Details may not add to totals due to rounding.

Appendix A. History of Perkins Acts

The following is a history of the legislation providing general federal support for career and technical education (CTE) programs that was enacted and eventually superseded by subsequent legislation until Perkins V.

The federal government has a long history of supporting workforce development, which includes CTE. The First Morrill Act of 1862 (7 U.S.C. §301 et. seq.) supported the development of the current system of land-grant colleges to teach the agricultural and mechanical arts to what it referred to as the "industrial classes." Federal funding for vocational education was initiated with the passing of the Smith-Hughes Act in 1917.[71] In 1917, vocational education was termed "vital to national defense and prosperity."[72]

Vocational Education Act of 1963 and Vocational Education Amendments of 1968

Though the federal government has played a role in CTE since the First Morrill Act of 1862 and the Second Morrill Act of 1890 (7 U.S.C. §312 et seq.), the Vocational Education Act of 1963 (VEA; P.L. 88-210) and the subsequent Vocational Education Amendments of 1968 (VE Amendments; P.L. 90-576), greatly expanded the federal role in vocational education, providing a modern precursor to Perkins V.

Whereas the Smith-Hughes Act focused on secondary schools, the VEA substantially increased funding for postsecondary and adult vocational education programs, and it incentivized the creation and expansion of vocational programs to serve at-risk students.[73] The VEA established what are now called Basic State Grants (BSG), which accounted for 90% of VEA appropriations. The VEA also increased support for dedicated vocational education schools, including providing funds for their construction. Additionally, the VEA provided funding for research, program development, and demonstration programs related to vocational education.

Within a year of the enactment of the VEA, federal appropriations for vocational education were five times greater than under the Smith-Hughes Act. The VE Amendments created a National Advisory Council on Vocational Education to advise the Commissioner of Education[74] on the administration and design of programs relating to vocational education and to collect and disseminate information about program effectiveness.

Perkins I

In 1984, the Carl D. Perkins Vocational Education Act (Perkins I; P.L. 98-524)[75] replaced the term *at-risk students* with *special populations* and defined them as students with disabilities,

[71] The Smith-Hughes Act was repealed by the Balanced Budget Act of 1997 (P.L. 105-33).

[72] U.S. Congress, House of Representatives, *Annual Report of the Federal Board for Vocational Education*, 65th Cong., 2nd sess., Doc. No. 16 (Washington, DC: GPO, 1917).

[73] The term *at-risk students* was defined in 1963 as "persons who have academic, socio-economic, or other handicaps that prevent them from succeeding in the regular vocational education program." The percentage of funding specifically designated for at-risk students was left undefined.

[74] Now the Secretary of Education.

[75] With this act, the VEA was renamed in honor of Carl D. Perkins (1912-1984), who was chairman of the House Committee on Education and Labor from 1967 to 1984. The Perkins Federal Student Loan Program, while authorized by entirely different acts, is also named after Carl D. Perkins, leading to occasional confusion between the two programs, though Perkins loans have not been disbursed since 2018.

economically disadvantaged students, adult students, single parents and homemakers, and students in correctional institutions.[76] Perkins I required each state to spend 57% of its BSG funds on activities targeting the vocational education of special populations. The remainder was to be spent on the improvement of vocational education programs.

Perkins II

The Carl D. Perkins Vocational and Applied Technology Education Act Amendments of 1990 (Perkins II; P.L. 101-392) made several revisions to the Perkins programs. The act established the Tech Prep program,[77] which was designed to coordinate secondary and postsecondary vocational education activities into a coherent sequence of courses. The law also required that at least 75% of BSG funds be allocated to local recipients and eliminated most of the set-asides (formerly 57%) for special populations from Perkins I. Additionally, Perkins II required states to develop and implement performance standards and measures such as program completion and job placement rates.

Perkins III

The Carl D. Perkins Vocational and Technical Education Act of 1998 (Perkins III; P.L. 105-332) increased the share of BSG state funds distributed to the local level to 85%, of which up to 8.5% could be reserved for programs in rural and other high-need areas. Perkins III also expanded state accountability, introducing core indicators of performance and adjusted levels of performance on each core indicator to be negotiated between the state and the Secretary of Education. Sanctions based on states failing to meet performance levels were also introduced, as were incentive grants to states for exceeding performance levels.

Perkins IV

The Carl D. Perkins Career and Technical Education Improvement Act of 2006 (Perkins IV; P.L. 109-270) replaced the term *vocational and technical* with *career and technical*, introduced the specific program of study requirement, and created an explicit linkage between CTE and academic standards.[78] State plans had to include accountability indicators and performance levels negotiated with the Secretary of Education.[79]

[76] The term *special populations* is currently defined in Section 3(29) of Perkins IV and means individuals with disabilities; individuals from economically disadvantaged families, including low-income youth and adults; individuals preparing for non-traditional fields; single parents, including pregnant women; out-of-workforce individuals; English learners; homeless individuals; youth who are in, or have aged out of, the foster care system; and youth with a parent who is a member of the Armed Forces or is on active duty.

[77] The Perkins V reauthorization eliminated the Tech Prep program.

[78] For more information on differences between Perkins IV and Perkins V, see *Reauthorization of the Perkins Act in the 115th Congress: The Strengthening Career and Technical Education for the 21st Century Act*.

[79] The Perkins V reauthorization removed the required negotiations between states and the Secretary on levels of performance.

Appendix B. Basic State Grants Allocation Formula

This appendix provides a detailed description of the formula used for allocating Perkins V Basic State Grant (BSG) funds to the states after set-asides for the outlying areas, the Native American CTE Program, and the Native Hawaiian CTE Program have been reserved.

Foundational Grants

After reservations for Guam, American Samoa, the Commonwealth of the Northern Mariana Islands, the Republic of Palau, and the Native American and Native Hawaiian CTE Programs, BSG funds are awarded to the 50 states, the District of Columbia (DC), Puerto Rico, and the U.S. Virgin Islands. Each of the 50 states, DC, Puerto Rico, and the U.S. Virgin Islands receives a foundational grant equal to its FY2018 allocation. If the appropriation falls below the FY2018 appropriation, foundational grant allocations are ratably reduced.

New Money Allocations

When total appropriations exceed FY2018 appropriations, the formula first allots up to one-third of the new money to bring states[80] with initial allocations below 0.5% of total allocations up to the 0.5% level.[81] States that are furthest below the 0.5% mark would receive proportionately larger shares of the new money than states that are closer to, but still below, the 0.5% minimum. If total appropriations are sufficiently high, then each state would be able to reach the 0.5% minimum allocation. States benefitting from this use of the new money are often referred to as the *small states*.

The remaining funds—at least two-thirds of the new money, plus any funds left over from bringing the small states up to the 0.5% minimum—would be allocated to all states based on their population and per capita income factors. The formula for state allocations was historically designed to target funds toward states with higher proportions of 15- to 19-year-olds. States with lower per capita incomes are also targeted by the formula.

The formula depends on three population groups in each state: persons aged 15 to 19, 20 to 24, and 25 to 65. Each of these populations is then weighted in the calculation of initial allocations. The largest weight (0.5) is assigned to the 15 to 19 age group. The 20 to 24 age group is weighted 0.2, while the 25 to 65 age group is weighted 0.15. A final calculation, based on the state's proportion of the total funding determined by the sum of the first three calculations, is weighted 0.15.

These population-based allotments are adjusted based on states' PCI, with the allotment ratios bounded between 0.4 (for states with PCIs substantially higher than the national average) and 0.6 (for states with PCIs significantly lower than the national average). In accordance with statutory provisions, a state's PCI is defined as that state's average PCI over the most recent three years for which data are available.

[80] *States* includes DC, Puerto Rico, and the U.S. Virgin Islands.

[81] The new money would not be used to bring the U.S. Virgin Islands' allocation to the 0.5% level, according to Section 111(a)(4)(D) of Perkins V.

The second factor in the formula is the state's allotment ratio. The ratio is calculated by dividing a state's PCI[82] by the national average PCI. That quotient is then multiplied by 0.5 and subtracted from 1.

$$Allotment\ Ratio = 1 - (0.5 \times \frac{State\ PCI}{National\ PCI})$$

This calculation results in higher allotment ratio values for states with low PCIs and lower allotment ratio values for states with high PCIs. No state may have an allotment ratio higher than 0.6 or lower than 0.4.[83] Allotment ratios for states with higher or lower calculated ratios are set to either 0.6 or 0.4, respectively.

Each state's new money allocation is calculated by adding four quantities based on weighted population-based calculations focused on three age groups (three quantities are derived through a separate calculation for each age group, and one is based on summing total funding determined through these separate calculations for each of the age groups). As is delineated below, the PCI-based allotment ratios are applied to these calculations.

$$New\ Money\ Allocation = \left(Allocation_{15-19} + Allocation_{20-24} + Allocation_{25-65} + Allocation_{prop}\right) \times (Total\ Appropriations - set\ asides - small\ states\ money)$$

The first three components of the sum are calculated by multiplying a state's PCI-based allotment ratio by its population in the given age group, and then dividing the result by the sum of all such products across all the states. That number is then multiplied by the weight corresponding to the particular age group:

$$Allocation_{15-19} = 0.5 \times \frac{State\ Allotment\ Ratio \times State\ Population_{15-19}}{Sum\ of\ (State\ Allotment\ Ratio \times State\ Population_{15-19})\ for\ all\ states}$$

where 0.5 is the weight factor associated with the 15 to 19 population group.

Correspondingly, the allocations associated with the 20 to 24 and 25 to 65 population groups are calculated as follows:

$$Allocation_{20-24} = 0.2 \times \frac{State\ Allotment\ Ratio \times State\ Population_{20-24}}{Sum\ of\ (State\ Allotment\ Ratio \times State\ Population_{20-24})\ for\ all\ states}$$

where 0.2 is the weight factor associated with the age 20 to 24 population group, and

$$Allocation_{25-65} = 0.15 \times \frac{State\ Allotment\ Ratio \times State\ Population_{25-65}}{Sum\ of\ (State\ Allotment\ Ratio \times State\ Population_{25-65})\ for\ all\ states},$$

where 0.15 is the weight factor associated with the 25 to 65 population group.

[82] For the purposes of this formula, a state's per capita income is defined as the average of that state's per capita income values for which data are available. A state's per capita income for a given year is calculated by dividing the total personal income in that state by the state's population.

[83] The U.S. Virgin Islands and Puerto Rico are automatically assigned allotment ratios of 0.6.

The fourth component of a state's initial allocation, $Allocation_{prop}$ in the above formula, is calculated as that state's proportion of the total funding determined by the sum of the first three calculations. This last component is given a weight of 0.15. As a result, $Allocation_{prop}$ is calculated as follows:

$$Allocation_{prop} = 0.15 \times \frac{Allocation_{15-19} + Allocation_{20-24} + Allocation_{25-65}}{Sum\ of\ (Allocation_{15-19} + Allocation_{20-24} + Allocation_{25-65}) for\ all\ states}.$$

To recap, the largest proportion (50%) of each state's new money allocation under the formula is assigned based on the number of persons aged 15 to 19, which is the traditional target population for CTE programs. The inclusion of PCI in the formula helps states with lower PCIs receive a greater share of available funds. Thus, states with lower PCIs and higher numbers of persons aged 15 to 19 will get the highest new money allocations.

Appendix C. Basic State Grants Allocation Table

Table C-1. Basic State Grant Allocations, FY2020

(Dollars in thousands)

State	FY2020 Allocation
Alabama	21,697
Alaska	5,136
Arizona	30,789
Arkansas	13,246
California	128,798
Colorado	19,174
Connecticut	11,011
Delaware	5,557
District of Columbia	5,136
Florida	73,997
Georgia	45,925
Hawaii	6,247
Idaho	7,719
Illinois	45,434
Indiana	28,924
Iowa	12,870
Kansas	11,553
Kentucky	19,666
Louisiana	22,413
Maine	6,247
Maryland	18,227
Massachusetts	20,909
Michigan	41,898
Minnesota	18,999
Mississippi	14,329
Missouri	25,263
Montana	6,018
Nebraska	7,336
Nevada	11,435
New Hampshire	6,247
New Jersey	25,793
New Mexico	9,353
New York	57,681

State	FY2020 Allocation
North Carolina	43,398
North Dakota	5,136
Ohio	47,064
Oklahoma	16,291
Oregon	15,586
Pennsylvania	45,115
Rhode Island	6,247
South Carolina	21,780
South Dakota	5,247
Tennessee	27,353
Texas	113,620
Utah	15,314
Vermont	5,136
Virginia	28,777
Washington	23,278
West Virginia	8,953
Wisconsin	22,715
Wyoming	5,136
Puerto Rico	19,508
Virgin Islands	1,010
American Samoa	384
Northern Mariana Islands	384
Guam	724
Native American and Native Hawaiian Programs	16,032
Freely Associated States	176
Other	3,206
Total	**1,282,598**

Source: U.S. Department of Education, Budget Service, *State Tables by Program,* October 26, 2021, https://www2.ed.gov/about/overview/budget/statetables/index.html.

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